

APPLICATION OF COMPUTER SCIENCE AND ENGINEERING IN THE FIELD OF HEALTH AND MEDICAL SCIENCES

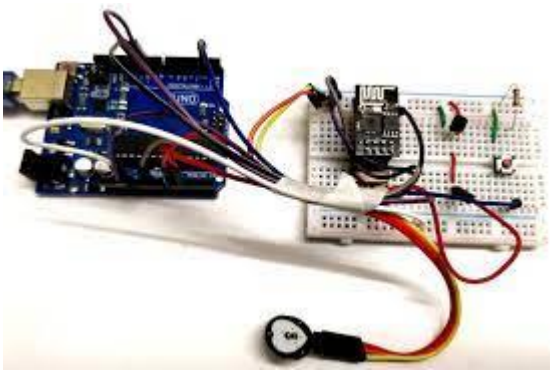
AUGMENTED REALITY IN MEDICAL IMAGING



Medical image processing is one of the applications of computer vision technology. 3D images obtained with the help augmented reality help radiologist to deliver a clear picture of the structural anomalies and provides a more accurate diagnosis. CT Scans with the help of AR give “real world view ‘this helps doctors actually see under the skin. This enable

doctors to know exactly where to inject and make proper incisions. Augmented Reality in tumor imaging provides insights on changes in tumor size as well as other details such as end-to-end structure, shape, and margin. It also provides 3D location of speculations and proximity to adjacent structures.

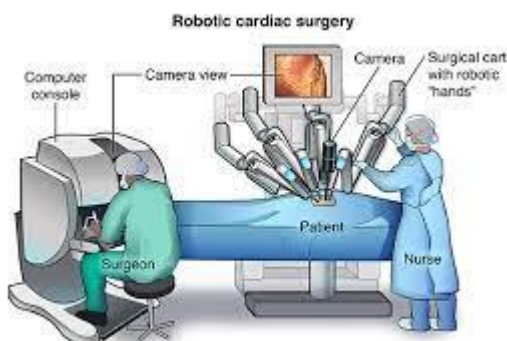
IOT BASED REMOTE PATIENT MONITORING



IOT has been widely used for Remote Patient Monitoring. Wearable devices with embedded sensors can monitor patient condition whole day long and can notify the doctor. With the Personal Healthcare Devices, an object in an IOT system medical staffs can monitor their patients at home. Sensors trace patient health if they find any

changes patient BP or pulse rate, the system automatically sends an alert to the doctor regarding the patient status over IoT and shows the details of BP, heartbeat and temperature of patient.

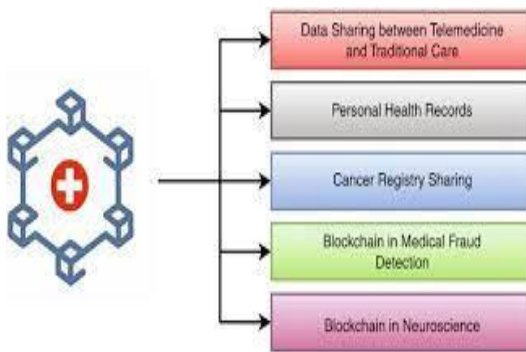
ROBOTICALLY-ASSISSTING SURGERY



Surgical procedures planning, teaching and performing of many surgical procedures are assisted by computers. This helps surgeons to align patients’ bones and joint implants with more accuracy. Computers offers visual mapping that helps doctors to take crucial decisions during orthopedic surgeries. Recent development in field of surgery is robotically-assisted surgery (RAS). RAS allows surgeons to use robotic devices and computer software to complete minimally-invasive

procedures. In certain parts of the body robot device can complete complex operation with the guided instructions of surgeon.

BLOCKCHAIN IN TELEMEDICINE



Telemedicine is “Computers for remote patient”. Patients far away or in rural areas can communicate with doctors with the help of computers or smart phones. Thus telemedicine offers an opportunity for medical institutions to start early treatment of faraway. During any natural disaster areas, telemedicine can greatly improve care. But there is a risk of patient data being leaked because of absence of security controls over the collection, use and transfer of data. With

Block chain in telemedicine facilitate a more efficient way to securely store and transfer data within a telehealth system, as well as sharing of data across healthcare organizations. Medical records that contains large amounts of information and data and with the help of blockchain, are stored in secure, fragmented systems enabling providers to store complete patient history and securely encrypt medical data.

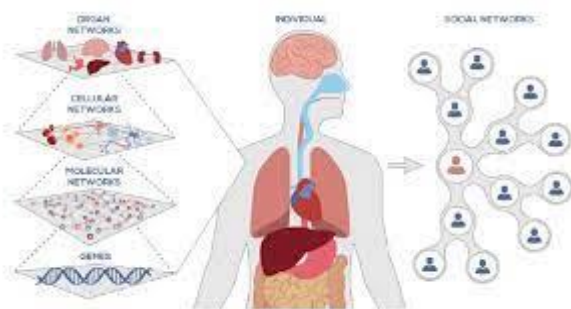
DATA ANALYSIS IN MEDICINE



In medical research large number of data is collected. This data is to be compiled, analyzed and interpreted. For this purpose, certain statistical methods are to be applied, these include calculation of standard deviation, standard error, application of tests of statistical significance like Z Test, unpaired and paired t test and chi-square test. Statistical

methods are time consuming. With the help of computer, large number of statistical calculations can be performed in a very short time.

COMPUTATIONAL MODELLING



Computational modeling in health care can be used to create simulations of complex biological mechanisms and predict outcomes. Popular computational modeling applications in medicine for premeds include tracking infectious diseases, predicting drug side effects and calculating the

effects of public health interventions. Most recently, computational models have been applied to COVID-19 outbreak predictions.

DIGITAL HEALTH.



health apps.

Several start-up companies and other organizations are building health apps, ranging from chronic disease management to patient scheduling to helping clinicians write their clinical notes. Premeds can join digital health companies or build their own

MEDICAL DECISION MAKING.

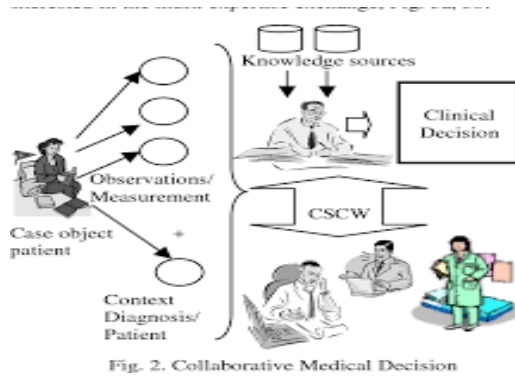


Fig. 2. Collaborative Medical Decision

Physicians and premeds can use computer software and algorithms to help with medical decision-making. For example, based on clinical findings and lab data collected, doctors can be prompted to order additional tests for their patients.

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